

Wavin Standard Maintenance Recommendation Letter

(For AquaCell® Geocellular Stormwater Systems – U.S. & Canada)

To Whom It May Concern,

Wavin's AquaCell® geocellular stormwater management systems are designed as modular, subsurface structures that provide long-term stormwater storage, detention, and infiltration. These systems require **minimal routine maintenance** when properly installed with appropriate upstream pretreatment.

Under **normal operating conditions**, AquaCell® systems should be maintained according to the following recommended schedule:

Recommended Maintenance Frequency

- **Annual inspection and maintenance** of the system and associated pretreatment structures.
- This typically includes checking upstream catch basins, sediment traps, filter media, and inlet/outlet structures, and cleaning them as needed to prevent debris from entering the tank.
- Semi-annual inspection of the AquaCell® is recommended through the first (1st) year of operation.
 - Inspection can be made via cleanout access points located above inflow/outflow pipes (if required by application specifications).
 - For maintenance & inspection guidance, reference **AquaCell® O&M Manual**
- Additional maintenance should only be required following unusual events such as heavy construction runoff, extreme sediment loading, or spill incidents.

System Maintenance Characteristics

AquaCell® units are made from durable, inert polypropylene and contain no moving parts. The structures are **not designed to require monthly or high-frequency internal cleaning**, provided that typical stormwater pretreatment components are used and maintained.

Local Regulatory Guidance

Stormwater regulations can vary between jurisdictions.

Wavin recommends that landowners and operators always refer to local stormwater regulation guidelines for specific inspection and maintenance requirements pertaining to underground geocellular stormwater systems.

If you have any additional questions or need job-specific documentation, Wavin's Technical Support team is available to assist.

Sincerely,

Wavin Technical Support

Orbia Building & Infrastructure

Overview of Maintenance for AquaCell System

The Wavin AquaCell is an innovative stormwater management system designed for the efficient collection, retention, and infiltration of rainwater. It consists of modular cells that can be configured to meet various project requirements, providing a flexible and sustainable solution for managing stormwater runoff. Maintenance of the Wavin AquaCell is essential to ensure its efficient and reliable operation. Regular maintenance involves cleaning the system to remove debris, sediment, and other obstructions that can cause blockages and reduce the system's capacity. This process helps prevent flooding, minimizes environmental impact, and extends the lifespan of the infrastructure. By conducting routine inspections and cleaning, potential issues can be identified and addressed before they escalate into major problems, ensuring the Wavin AquaCell system continues to protect communities and properties from stormwater-related damage.

A. Preparation & Planning

1. Inspect the AQUACELL system

- Conduct a preliminary inspection to identify areas that require cleaning and assess the condition of the system.

2. Gather Necessary Equipment

- Ensure you have all the necessary equipment, including a high-pressure water jet, vacuum truck, personal protective equipment (PPE), gas detector, and communication devices.

3. Safety Protocols

- Establish safety protocols, including confined space entry procedures, traffic control measures, and emergency response plans.



B. Safety Measures

1. PPE

- Equip all workers with appropriate PPE, including hard hats, gloves, safety goggles, respirators, and protective clothing.

2. Gas Detection

- Use gas detectors to check for hazardous gases like methane, hydrogen sulfide, and carbon monoxide in the system before entry.

3. Ventilation

- Ensure proper ventilation to prevent the accumulation of hazardous gases.

4. Confined Space Entry

- Follow confined space entry procedures, including having a trained attendant and using a harness and lifeline for the worker entering the space.



C. System Isolation

1. Isolate the Area

2. Block off the section of the stormwater system being cleaned to prevent water from entering during the cleaning process.

3. Notify Relevant Parties

- Inform local authorities, businesses, and residents about the cleaning operation and potential disruptions.



D. Debris Removal

1. **Access the AQUACELL through 10" Ø ventilation / inspection ports located directly above the system.**
2. **Initial Debris Removal**
 - a. Use manual tools or mechanical equipment to remove large debris and objects that may obstruct the cleaning process.
3. **High-Pressure Water Jetting**
 - a. Use a high-pressure water jet to dislodge sediment, silt, and smaller debris from the walls and floor of the system.



E. Vacuum Extraction

1. **Vacuum Truck**
 - a. Use a vacuum truck to suck out the dislodged debris and water from the system. Ensure the vacuum hose reaches all areas of the section being cleaned.
2. **Repeat Process**
 - a. Repeat the jetting and vacuuming process until the section is thoroughly cleaned.

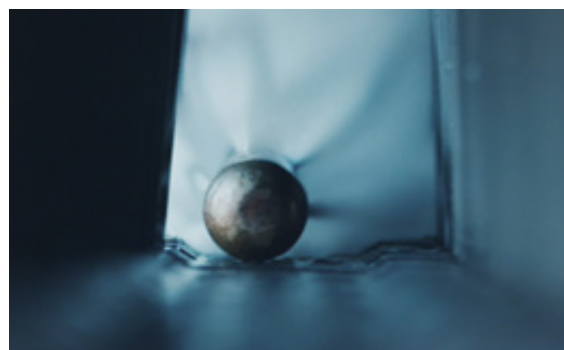


F. Inspection and Verification

1. **Inspect Cleaned Sections**
 - a. Conduct a post-cleaning inspection to verify that all debris has been removed and the system is functioning properly.
2. **CCTV Inspection**
 - a. Use a CCTV camera to inspect the interior of the system for any remaining debris, blockages, or structural issues.

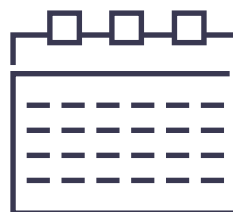
G. Final Steps

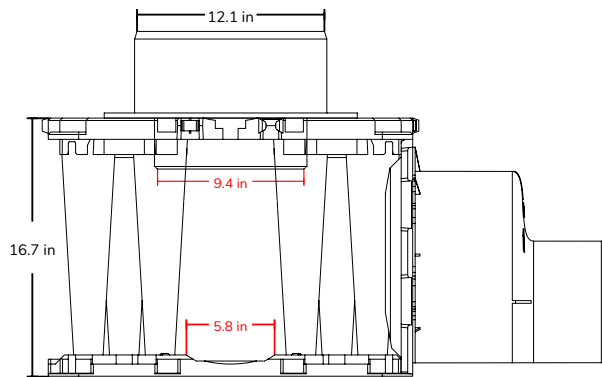
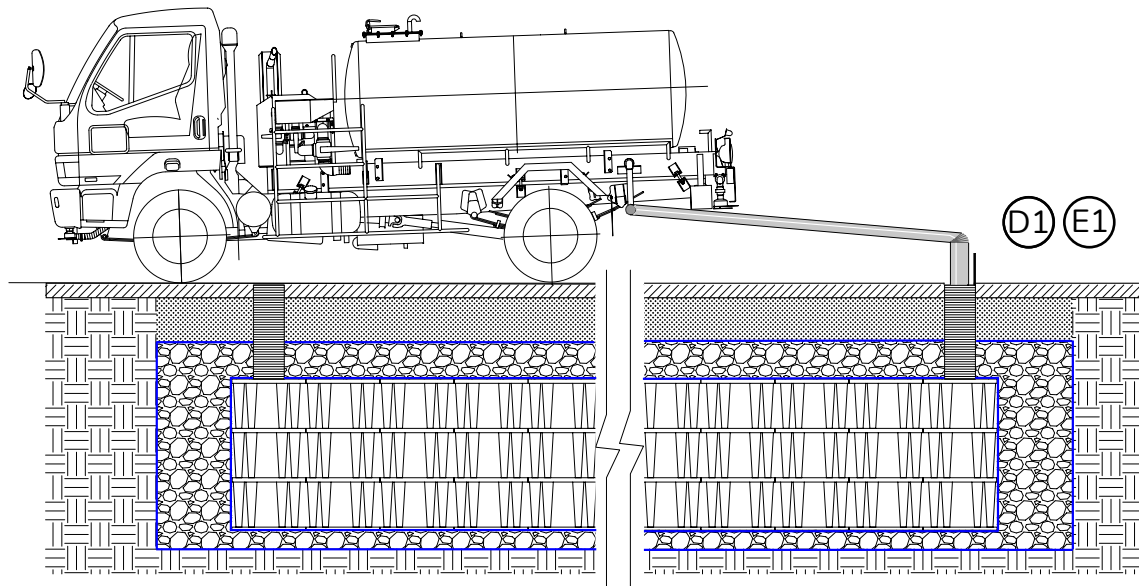
1. **Dispose of Waste**
 - a. Properly dispose of the collected debris and wastewater according to local regulations and environmental guidelines.
2. **Document the Process**
 - a. Record the cleaning process, including areas cleaned, equipment used, and any issues encountered.
3. **Reopen System**
 - a. Remove isolation measures and restore the AQUACELL system to normal operation.



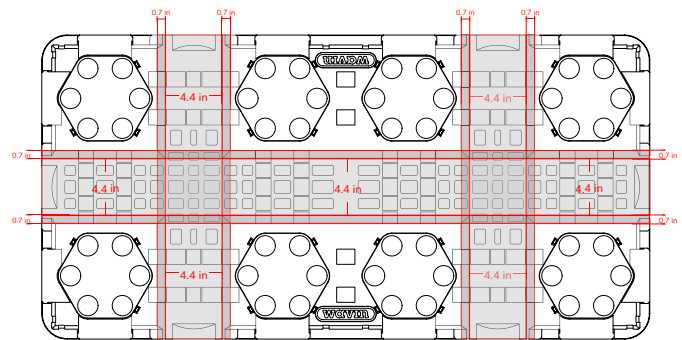
H. Regular Maintenance

1. **Schedule Regular Cleanings**
 - a. Establish a regular maintenance schedule based on the system's usage and local conditions to prevent future blockages and ensure efficient operation.
2. **Monitor System Performance**
 - a. Continuously monitor the system's performance and address any issues promptly to maintain its effectiveness.





9.4" open entry through top adapter and 5.8" width of inner channels between columns allow for clear access into interior of AQUACELL tank



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